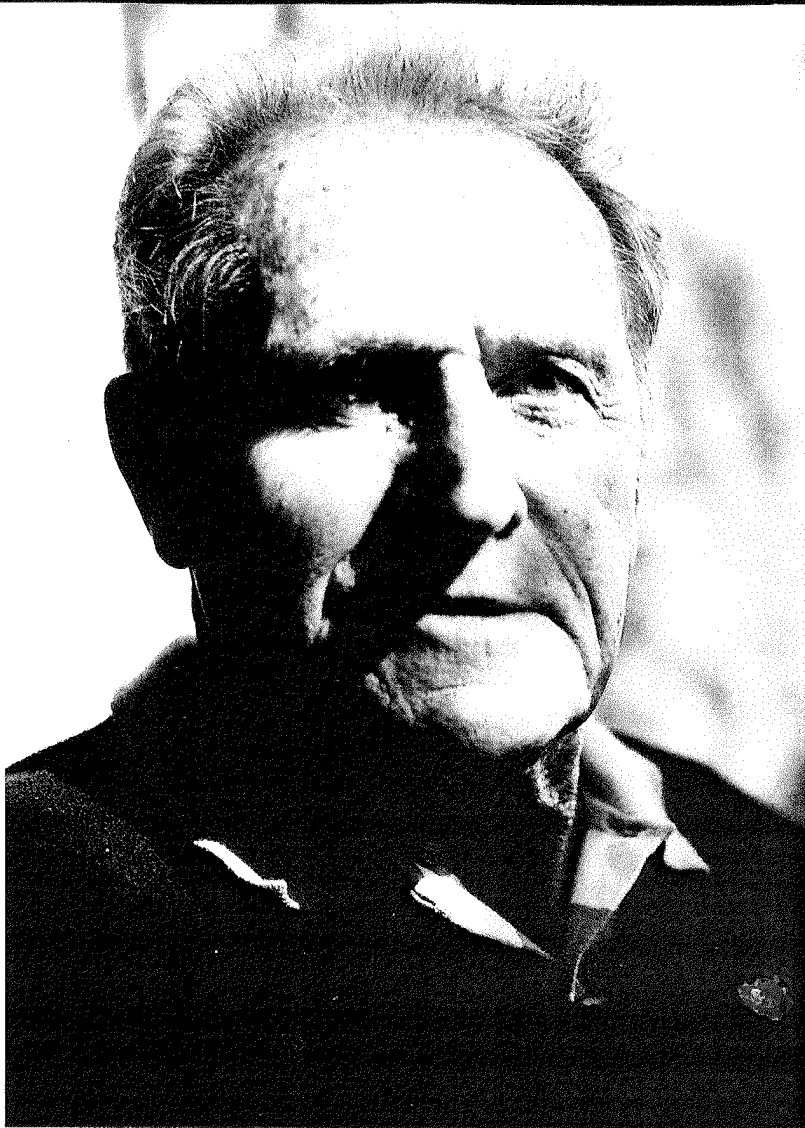




ESSENTIAL LYDIARD

PRINCIPLES, NOT FORMULAS, ARE THE KEYS TO SUCCESSFUL TRAINING

BY LORRAINE MOLLER

There's a story told about Arthur Lydiard and his golden boys at the '64 Olympics in Tokyo. Down at the training track, with their rivals looking on, they ran an impressive interval session of 20 quarters. The next day one of these opponents, a talented interval-trained Canadian in his first Olympics, showed up at the track with his coach and proceeded to run the same session. Lydiard's boys cheered him on as he ran each interval faster than the last. When it was all done a reporter asked Lydiard what he thought of the kid's workout. "I think it was the last nail in his coffin," Lydiard replied.

"But your boys ran the same session yesterday."

"Yes, but my boys needed it."

Perfectly peaked, Lydiard's protégé, Peter Snell, won two gold medals.

His teammate, John Davies, won a bronze. The Canadian who had eclipsed the Kiwis' training run with his own failed to advance to the finals in his event — as Lydiard had predicted.

One of my former coaches, Ron Daws, often quoted the above story to illustrate his axiom: "Good training and bad training look exactly the same on paper." Twenty quarters can bring one runner to his or her peak and bury another in a hole. Daws, adhering to Lydiard's philosophy, recognized that stand-alone workouts mean little; it is their contextual application to the advancement of the athlete's goal that matters. Training, then, is not a series of numbers that can be universally applied but is rather the art of combining measure, timing and sequence to the specific needs of the individual. Herein lies the brilliance of the coaching of Arthur Lydiard.

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Photo: Lydiard Foundation

My first coach, John Davies, the same Lydiard protégé who won bronze in Tokyo, called Arthur to ask him how to adapt the training for a promising 14-year-old girl. He didn't want to wreck me. Arthur advised him to build a base by increasing my mileage. I began on regular runs, soon reaching my target of 40 miles per week, which I continued for some months. There were never any hard and fast rules that are regularly attributed to Lydiard, such as the 100-miles-per-week mandate, but rather a careful consideration for my age, gender, ability, fitness and event. And that's how it was for the next 28 years of my career as an international athlete under the guidance of three consecutive Lydiard coaches: Davies, Ron Daws and Dick Quax. My schedules were never replicated in content; however, the principles that defined them as good Lydiard training (and outlined in this article) were unwavering.

01

MILES IN THE BANK

MAXIMIZE AEROBIC CAPACITY FIRST

The first phase of Lydiard training is endurance/aerobic development. Think of aerobic running as home base — the place where we hang out until we are mature enough to leave, and the place we always come back to for rest and recuperation. As the miles stack up, we increase the capacity of both the heart and the lungs for work, build our circulatory network to the muscles through increased capillarization, increase the number of mitochondria in the muscle cells, and develop other beneficial metabolic and enzymatic pathways for gathering and converting oxygen to energy. Once these structures are established they allow us to respond and recover quickly. Aerobic training, then, is training for all other types of training.

Lydiard often described miles as money in the bank — the more you have the greater your currency to buy ATP's (the units of energy your muscles need for contraction) and the faster you will eventually be able to race in any event that has a large aerobic component. From the 800 meters on up the body's energy needs are met primarily from aerobic metabolism, with any race over 5,000m more than 85 percent dependent on aerobic ability.

There is no argument: The greatest gain for any aspiring distance athlete is made by spending the majority of time on aerobic development. And for those recreational athletes training for a weekend road race, this is the only phase of Lydiard training that is initially indispensable.

THE KEY TO TRAINING IS TO APPLY APPROPRIATE WORKLOAD
AND RECOVERY SO THAT YOU GAIN STEADY CONTINUOUS IMPROVEMENT.

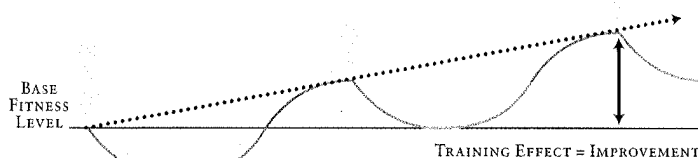


FIGURE 1:
BASIC PRINCIPLE OF ADAPTATION

While easy running is always the safest place to start, it is often the case that the cardiovascular system develops quickly while the musculoskeletal system tends to lag somewhat. The golden rule is that you can never progress faster than your slowest part will allow. Case in point: For the past 20 years Japanese teams have regularly altitude-trained in Colorado. I often recognized some of these petite women, backpacks on, hiking up the steep incline of Gold Hill some 15 miles out of Boulder. Taking the base-building strategy one step further, the coach had prescribed extended hiking and jogging sessions during year one of the four-year Olympic cycle solely to condition the musculoskeletal system for base-building. With strong tendons and ligaments these runners could then handle up to 200-mile training weeks without injury, and thus develop a superior aerobic capacity.

02

FEELING-BASED TRAINING

TUNE IN TO YOUR INNER COACH

One of the benefits of the buildup phase is that these longer aerobic runs allow you to safely build a rapport with your body. This is a greatly underrated aspect of training, probably because feelings have generally not been a part of any serious exercise discussion. But being able to precisely gauge one's effort over time is an ability that is the hallmark of all great athletes. They can run the razor's edge, knowing how to pitch their effort and energy to extract the best from their bodies on any given day.

Lydiard fostered this ability in all his runners

by prescribing feeling-based training goals. A typical running schedule would ask for half effort on one day and three-quarters or seven-eighths effort on another. Half-effort was half the effort of a full effort, and so on. In the days before heart rate monitors and chronographs one simply had to guesstimate until they got the gist of it, which they did quickly.

Modern technology can be helpful training wheels to developing this rapport but an overreliance on outside feedback, whether a beeping monitor or a coach bellowing splits from the sidelines, is also a dangerous trap. Ultimately we are on our own in competition, reliant on the clarity of communication between our mind and our body. I call this the "inner coach," the voice within that knows exactly what we need to do at any point in time to reach our potential. Whether it tells us to back off, relax, pick it up, or make a break, in hindsight it was always the right thing to do. As we learn to trust the inner coach over time, the stronger its voice becomes.

03

RESPONSE-REGULATED RECOVERY

BALANCING WORKOUTS

Training can be defined as specific stress applied to the body to invoke a corresponding adaptation. The training stimulus (workout) causes a temporary breakdown in the body (catabolic phase) followed by the adaptive period (recovery), during which the body rebuilds itself so as to better withstand the stress that it has just endured (anabolic phase). Breakdown and buildup better than before, breakdown and buildup better than before; this is the rhythm of good training. Interestingly, the desired training effect does not take place during the workout but during the recovery. We improve not while we are training but while we are resting. More often than not, bad training is a mismatch of breakdown and buildup: either the workout is too hard or the recovery is inadequate, or both. (Rarely among motivated athletes is it the result of under-training.)

The tendency for many runners to overtrain and/or under-recover is underscored by an inflexible training schedule. While Lydiard pushed his runners, he offset the overtraining

syndrome by preparing them for optimal recovery with base training, gearing the training to be feeling-based, and adjusting workouts according to the athlete's recovery response. The art of good training calls for an accurate assessment of which side of the adaptation curve the runner is on — catabolic or anabolic — and prescribing appropriately: a recovery run or a workout.

There are simple ways of assessing this: An elevated morning heart rate, poor sleep, low energy, sore muscles and bad mood are all indicators that the runner needs further recovery and a workout of any intensity is contraindicated. Once the "spark" has returned the runner is ready for the next "stress."

Shigeharu Watanabe, coach of Yoko Shibui, whose personal best of 2:19 ranks her No. 7 all-time among women, served his coaching apprenticeship with Lydiard in New Zealand. Of his team he says, "We do only that which is necessary to do, we do not follow any numbers."

04

SEQUENTIAL DEVELOPMENT OF ENERGY SYSTEMS

A study performed by Dr. Izumi Tabata (and colleagues) at the National Institute of Fitness and Sports in Tokyo, Japan, showed that over a six-week period, interval training is more efficient than steady exercise of lesser intensity. This study has been quoted by get-fit-quick advocates as proof of the superiority of interval training. What they do not tell you is that anaerobic development takes four to six weeks to top out, and that when preceded by endurance and strength training the possibility of injury is decreased and the quality of the intervals is enhanced.

Whenever Lydiard was asked which part of the training was the most valuable, he always answered simply, "Everything." Often he would follow up with the question, "Who would want to eat a cake half-baked?" This holistic approach distinguishes Lydiard training from one-brand-fits-all formulas and squelches the quest for the "magic workout." There is no single element that makes one a champion. Rather, it is the progressive development of each energy system from

low intensity to high intensity, each stage preparing the runner for what is to come next. Speed is built on interval training which is built on strength which is built on endurance. When the runner finally reaches her target

with three Olympic gold medals and numerous world records to his name. In his post-athlete career he came to live in the U.S., where he became a doctor of exercise physiology and associate professor of internal medicine. With the eminent

credentials of both athlete and academic, who is better equipped to offer a final thought on the relevance of Lydiard today?

SNELL SAYS:

The core of Lydiard training is the quantity and quality of the base training. Although Arthur appeared to be very precise about the track training leading up to the racing season, I believe this was not as important as the base. Long, moderate-pace running is anabolic whereas

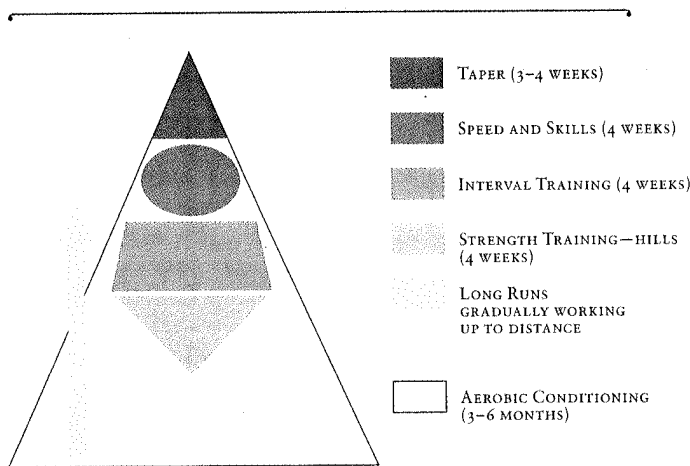
high-intensity

demanding training, while having its place, is catabolic. Thus the base is critical to prevention of overtraining.

Marathon training for an 800m runner is difficult for many coaches and particularly scientists to understand. Many have been quite dismissive about the benefits in the face of the outstanding results associated with marathon training. To them it makes no sense training slowly for a speed event. The rule of specificity is violated. Why then does it work? Today our knowledge of physiology provides some answers:

- Long endurance runs appear to provide protection against overtraining from too much high-intensity speed work. Therefore more race-related training may be accomplished.
- Activation of fast-twitch muscle fibers is normally accomplished by high-intensity interval runs. We now know that long moderate-pace runs also activate fast-twitch muscle fibers, after slow-twitch fibers have become glycogen depleted after the first one to two hours.

In sum, and in light of current physiology, there is little I would do differently today other than incorporate some long, easy intervals once a week during the base training. The training of current top athletes is testimony to the relevance of Lydiard training today. *WT*



THE BIGGER THE BASE, THE HIGHER THE PEAK.

FIGURE 2:
THE LYDIARD TRAINING PYRAMID

race, she possesses the entire gamut of adaptive responses from jogging to sprinting.

05

CORRECT TIMING

Peaking is a matter of correct timing. It is one thing to maximize the amount of energy at the disposal of an athlete. It is another to channel that energy into the event that matters. To ensure one's best form is achieved on competition day, a Lydiard schedule is always written from the goal backwards, allotting the amount of time needed for each phase and using the remaining time for base training. Thus the training pyramid serves the twofold purpose of physiological and mental focus to arrive at one point in time — the goal. There is no need for a separate mental training program of affirmation and visualization — it is built into the training program. As the athlete gets closer and closer to the top of the pyramid, the training increasingly simulates the race. There is nothing more confidence-building than the somatic knowing of thorough preparation.

Using the five principles of miles in the bank, feeling-based training, response-regulated recovery, sequential development of the various energy systems and correct timing, the what, when, how and why of every action from sleeping to running is another brick in the pyramid that takes one to the top.

Lydiard's greatest protégé, Peter Snell, retired



For a podcast with Lorraine Moller on Arthur Lydiard's key training principles, go to runningtimes/nov09.

HOW PETER SNELL TRAINED FOR A WR 1:44.3 ON GRASS IN 1962

LONG RUNS, HILL CIRCUITS, VERY VARIED TRACK WORK, MORNING RUNS

IN ABOUT AUGUST 1961, PETER SNELL RETURNED

from a successful European track tour exhausted from the frenetic racing and traveling, and didn't feel inspired to recommence a buildup until September 17. He established a twice-a-day routine of running to and from his work, and by October 21 won an exhibition mile race in 4:13, in the midst of accumulating the first 100-mile week of that buildup. By November, he was able to cover the 22-mile Waiatarua circuit confidently, covering 4 miles to Lydiard's home before running the full course in 2:11, his best since before his Rome Olympic 800m triumph.

The following day, he started two weeks of hill circuit training, with Wednesdays given over to club races, presumably over sprint and middle distances. After the first week of circuits, he ran a picnic meeting mile in 4:14. He continued his long Sunday runs, as well as morning runs of 10 miles during the week. At the end of his second week of hill circuit training, he competed in the Owairaka Marathon. He stayed with the leaders in a top field, being fourth at 20 miles. Half a mile later he was reduced to a walk and at 24 miles, he sat down.

He dredged himself to a very tired finish in 2:41. Later that day, he played in a social cricket match, and when he batted "lasted about three balls before being clean bowled ... my vision was obscured ... there was no coordination at all." Obviously Snell had managed to totally deplete his glycogen reserves.

He recovered all week, running another Waiatarua circuit the next weekend half an hour slower than usual, then the next Monday started track work with a 4:10 picnic meeting mile and a last 440 yards of 58.6 seconds. Thereafter followed a variety of track work on several afternoons a week, with 10-mile morning runs every day and the weekly long run.

He recorded a 9:18 2-mile in his first week of track training, followed soon after by a three-quarter mile in 3:04.5 and for the next two weeks, in training or at picnic race meetings, a number of hard time trial efforts over distances between 440 yards and 3 miles.

On Christmas Day, he ran a 1:52/4:15 half mile/mile double. Four days later, he ran 10 half-mile efforts in 2:10 average, with half-mile recovery jogs. All the while, the 10-mile morning runs and long Sunday runs continued. In the next few weeks, he ran sessions including a 9:12 2-mile the day before 10 x 440 yards in an average of 59.8 seconds, a half mile in 1:51, and a windy 3 miles in 14:23 (equivalent to about 14:50 for 5,000m), a few days before a 440 yards in 50 seconds.

He then started to ease up the pressure for a few days, before winning an international race series 880 yards in 1:48.2, passing through 800m in 1:47.7. Although due to run another international half mile the next Wednesday, he felt strong enough to run his usual 22 miles on the Sunday. (Barry Magee says that Snell's long runs during the business end of his track season were slow and restorative, about a minute a mile slower than his best aerobic speeds achieved in his buildup.) The next day he ran half an hour in the morning, and 8 x

150 yards, working up to sprint speed in the closing stages of each.

On the Wednesday, he ran 880 yards in 1:47.1, passing through 800m in 1:46.3, tying his Olympic record time. The 10-mile morning runs continued.

A week later, on January 27, 1962, he ran a world mile record of 3:54.4 with a last 440 yards of 54.8. Eight months later he won the 880 yard/mile double in the Commonwealth Games. Thirty months later he won the 800m/1500m double in the Olympics.

SO WHAT?

Well, obviously regular long runs don't harm one too much. When we look at what Snell did while juggling training, working, and a racing tour around the country, we can see one certain fact: Snell ran his world 800m record within two months of completing his first and last competitive marathon.

WHAT WAS THE PHYSIOLOGY AGAIN?

Before his marathon he'd run a number of hill circuits over a fortnight, which re-stimulated his powerful type IIb fast-twitch fibers with plyometric input that reversed the normal "size principle" of muscle fiber recruitment (smallest to largest). He'd also redeveloped a substantial aerobic capacity with consecutive 22-mile runs.

In running a marathon at a constant pace, Snell ran to the limits of his slow-twitch fiber capacity and then exhausted the glycogen stores in his fast-twitch fibers to adapt. They were now quite possibly acting as type IIa fast-twitch fibers, with aerobic endurance characteristics as well as a very large anaerobic glycolytic potential.

This potential, of course, couldn't be realized until a variety of hard fast workouts and races capitalized on their emergence. The rest is history, and we can learn from history. While most modern coaches wouldn't recommend a full marathon so close to track racing, at least one great coach of the 1970s was paying attention. John Walker ran for over 20 miles at the head of New Zealand's tough Rotorua Marathon in April 1975. Then he jogged off the course, with his mission accomplished, amid quite a few ruffled marathoners' feathers. Why?

A couple of months later he was the first man to smash the 3:50 mile barrier. His coach, Arch Jelley, was a member of Owairaka Athletic Club and applied the Lydiard principles famously.

Walker broke the world mile record with 3:49.4 in 1975 after a period of training over the Waiatarua circuit with Kevin Ryan, the top New Zealand marathoner of the time. Eight years later, in 1983, he trained on the same circuit with marathoner Chris Pilone, and subsequently he ran his PB over the mile, 3:49.08, still the New Zealand record. RT



SNELL RAN 1:47.6 TO BEAT JAMAICA'S
GEORGE KERR IN THE 880Y AT THE 1962
COMMONWEALTH GAMES IN PERTH.

Photo: Getty Images/Allsport/Hulton/Archive



Excerpted with permission from *Healthy Intelligent Training* by Keith Livingstone, a book that, in the author's words, is about, "applying the principles pioneered by Lydiard over 50 years ago, in a modern context, with the hindsight of current physiology."